LESSON 4.9
ABSOLUTE VALUE – PART 2

LESSON INTRODUCTION

MA.6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.

LEARNING TARGETS

- I can find the absolute value of a value and interpret it within the context.

BENCHMARK COHERENCE

WORKING TOWARD

MA.7.NSO.2.1 Solve mathematical problems using multistep order of operations with rational numbers, including grouping symbols, whole-number exponents, and absolute value.

ESSENTIAL QUESTIONS

- What is the relationship between the absolute value of opposites?
- What represents zero in the real-world situation?
- How do you determine and interpret the absolute value in real-world situations?

LESSON NARRATIVE

Students extend their working definitions of absolute value from Lesson 8 to include sensemaking in real-world contexts. Spiraled practice for identifying absolute value and plotting on a number line is scaffolded throughout this lesson, but each activity also incorporates the interpretation of that absolute value within the given context.

COMPONENT SUMMARY

4.9.1 WARM-UP *(~5 MIN.)*

Select opposites to fill blanks in an absolute value sentence

4.9.2 GUIDED PRACTICE *(15 MIN.)*

Determine and interpret absolute value and opposites in real-world situations

4.9.3 COLLABORATION *(10 MIN.)*

Collaboratively determine and interpret absolute values and opposites in real-world situations

4.9.4 YOUR TURN! *(10 MIN.)*

More practice determining and interpreting absolute values and opposites in real-world situations

4.9.5 WRAP-UP *(~5 MIN.)*

Demonstrate understanding of lesson learning goals

© Accelerate Learning Inc. – All Rights Reserved

RESOURCES

GLOSSARY

Key Terms:

- Speed

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- Consider creating sentence stems and answer choices for question 2 in 4.9.3 Collaboration and 4.9.4 Your Turn!

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think the absolute values of negatives are negative.
- Students may believe larger negative numbers also hold larger value (i.e. -7 is greater than -6).

THINGS TO CONSIDER

- Interpreting the meaning of absolute values in context can be challenging. Review the contexts and interpretations prior to the lesson to ensure interpreting is modeled correctly with students.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 4.9.3 Collaboration, students initially complete the questions independently. Then students discuss their responses and strategies with a partner. These discussions help students gain confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

In 4.9.3 Collaboration, students represent absolute values with number lines, tables, and written descriptions. Students deepen their understanding as they make connections among these representations.

DIFFERENTIATE INSTRUCTION

Some students may be unfamiliar with particular real-world contexts and related terms. While these contexts have been incorporated throughout the unit, students who are not familiar with bank accounts may engage with terms like “overdraft” and allow those terms to become obstacles to learning. Consider offering vocabulary cards for students to reference either on their desks or on the board for quick definitions of each term.

Possible terms to include as vocabulary cards:

- 4.9.2 Guided Practice and 4.9.3 Collaboration – Celsius or sea level
- 4.9.4 Your Turn! – Sea level, elevation, balance, or overdrawn
- 4.9.5 Wrap-Up – Withdrawal, deposit, balance

© Accelerate Learning Inc. – All Rights Reserved

4.9.1 WARM-UP: ABSOLUTE VALUE

Component Overview: *Students select opposites to fill blanks in an absolute value sentence. Students discuss their strategy for determining the missing values with a partner. This component solidifies student understanding that opposites have the same absolute value.*

TEACHER MOVES

Think-Pair-Share

Students independently choose opposites and determine their distance from zero on a number line. Then students discuss their answers and strategies with a partner.

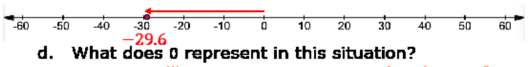
4.9.1 Warm-Up: Absolute value

1. Choose any value 1 through 9, along with its opposite, to complete the statement below:
   6 and −6 are 6 units away from 0 on the number line.
   Answers will vary. An example is filled in above.
2. Switch your answer with a partner to check.
3. Ask your partner their strategy for finding their answer and write their strategy below.
   Answers will vary.

© Accelerate Learning Inc. – All Rights Reserved

4.9.2 Guided Practice: Absolute Value and Opposites in the Real World

1. A tiger was rescued by The Brevard Zoo in Melbourne, Florida. The zoo veterinarian examined the tiger and determined the animal's weight to be 29.6 pounds below average.
   a. What value represents this situation?
      −29.6
   b. What is the absolute value of the value that represents the tiger's weight loss?
      $|-29.6| = 29.6$
   c. Model the tiger's weight loss on the number line.



   d. What does 0 represent in this situation?
      Answers will vary. Zero represents the change from average weight.
   e. If the tiger's weight had been 29.6 pounds above average instead of below, what would be the same about this situation?
      Answers will vary. The amount of change would remain the same.
2. Simone is doing research on different rides at the Strawberry Festival. She attaches an accelerator to a bumper car to measure the velocity of her car. She recorded the velocities, in miles per hour (mph), in the table.

Speed is the absolute value of velocity.

Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)
5	−3.1	20	−2.4	35	−5.2
10	−2.7	25	−6.3	40	3.1
15	4.8	30	2.7	45	−7.8

4.9.2 GUIDED PRACTICE: ABSOLUTE VALUE AND OPPOSITES IN THE REAL WORLD

Component Overview: *Students determine and interpret absolute values in real-world situations. The real-world contexts are related to changes in weight, speed, temperature, and elevation.*

ENRICHMENT

- When considering the changes in the tiger's weight, what does zero represent?
- How do you model a loss in weight?
- How do you model a gain in average?

TEACHER MOVES

Facilitate a conversation about speed and velocity. Provide examples and check for understanding to ensure students understand these terms.

In the context of velocity, the signs represent the direction the bumper car is headed.

 © Accelerate Learning Inc. – All Rights Reserved

4.9.2 GUIDED PRACTICE: ABSOLUTE VALUE AND OPPOSITES IN THE REAL WORLD (CONTINUED)

ENRICHMENT

Prompt students to reflect on how Bryan's friend might be more precise in his mathematical language.

- His friend uses the phrase "away from" – why could this have two possible meanings?
- How could this language be more precise to determine the actual temperature?

a. **Determine when the bumper car is at the same speed. List the times and the speed of the car at those times.**
At 5 seconds and 40 seconds, the car is traveling at 3.1 mph. At 10 seconds and 30 seconds, the car is traveling at 2.7 mph.

b. **At what time is the bumper car traveling at the slowest speed?**
20 seconds

c. **At what time is the bumper car traveling at the fastest speed?**
45 seconds

3. **Before Bryan goes snow skiing, he asks a friend about the weather. The friend says that the temperature is 5°C away from 0.**
Select all the temperatures that are possible.
- ☐ −10°C
- ☐ −5°C
- ☐ 0°C
- ☐ 5°C
- ☐ 10°C

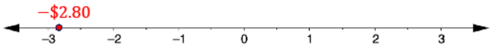
4. **A part of the city of New Orleans is 19 feet below sea level. We can use "−19 feet" to describe its elevation, and "|−19| feet" to describe its vertical distance from sea level. In the context of elevation, what do each of the following numbers describe?**

Value	Meaning in Context
25 feet	*The elevation is 25 feet above sea level.*
|25| feet	*The vertical distance of the elevation from sea level is 25 feet.*
−8 feet	*The elevation is 8 feet below sea level.*
|−8| feet	*The vertical distance of the elevation from sea level is 8 feet.*

© Accelerate Learning Inc. – All Rights Reserved

4.9.3 Collaboration: Absolute Value and Opposites in the Real World

1. Anthony owes his friend $2.80.
   a. Sketch a number line that represents this situation.



   b. Represent this situation as a rational number.
   −2.8
   c. What is the absolute value of the rational number?
   2.8
   d. What does 0 represent in this situation?
   Answers will vary. At 0, Anthony neither owes money nor has money.

2. With your partner, describe the meaning of each value in the context of temperature, where the freezing point is 0°C.

Temperature	Interpretation
1°C	*Answers will vary. 1 degree above zero (or above the freezing point).*
−4°C	*Answers will vary. 4 degrees below zero (or below the freezing point).*
$\lvert 12 \rvert$°C	*Answers will vary. A change of 12 degrees above zero (freezing).*
$\lvert -7 \rvert$°C	*Answers will vary. A change of 7 degrees below zero (freezing).*

4.9.3 COLLABORATION: ABSOLUTE VALUE AND OPPOSITES IN THE REAL WORLD

Component Overview: *Students collaboratively determine and interpret absolute values and opposites in real-world situations. Real-world situations include owing money and temperature changes. Students represent absolute values with numbers and written descriptions.*

TEACHER MOVES

Think-Pair-Share

Have students initially complete the questions independently. Then have students discuss their responses and strategies with a partner. Alternatively, students can collaborate and work on the activity in pairs or groups, followed by whole-class discussion.

ENRICHMENT

- What does zero represent in this situation?
- What does a positive value represent?
- What does a negative value represent?

© Accelerate Learning Inc. – All Rights Reserved

4.9.4 YOUR TURN!

Component Overview: *Students practice determining and interpreting absolute values and opposites in real-world situations. The real-world situations addressed include a bank account balance and elevation.*

ENRICHMENT

- What happens to your account balance when you spend money?
- What about when you deposit money?
- If your account balance is negative, what can be done to get out of the negative and back to zero?

4.9.4 Your Turn!

1. **The absolute value of the elevation of Enterprise, Florida, is 19.7 feet. In relation to sea level, write the two values that could represent the elevation of Enterprise, Florida.**
   19.7 feet or −19.7 feet

2. **Jasmine received a notice that her bank account was overdrawn, showing a balance of −\$20.48. What does the absolute value of her balance mean in this situation?**
   Answers will vary. The absolute value of her balance in this situation means that \$20.48 more was taken out of her account than she had in the account.

© Accelerate Learning Inc. – All Rights Reserved

4.9.5 Wrap-Up: Ticket Out the Door

1. The number -220 represents a withdrawal in dollars from a checking account, and 220 represents a deposit in dollars to the same checking account. What do the absolute values of these numbers mean in terms of this context?
   Answers will vary. The absolute value means that the change in the account balance is $220 in both situations.

2. A bank account has an account balance of $\$150$. For each description, check the value(s) of each expression that correctly matches.

Description	12	$\lvert -12 \rvert$	$\lvert 12 \rvert$	-12
a withdrawal of $\$12$				X
a deposit of $\$12$	X			
a change of $\$12$ above the balance			X	
a change of $\$12$ below the balance		X		

4.9.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *The Wrap-Up assesses students' abilities to interpret rational numbers and absolute values in real-world situations. Use data from this formative assessment to determine which students need remediation before the next lesson.*

TEACHER MOVES

Remind students to think of these changes in terms of the bank account. While withdrawing $12 does increase the money in their wallet, question 2 is all about the account amount. Likewise, the phrase "change of $12" indicates a direction or operation to apply to the balance.

© Accelerate Learning Inc. – All Rights Reserved